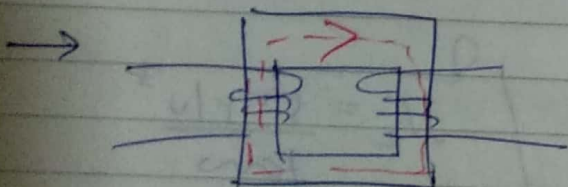


⇒ 3- ϕ Induction motor

Transformer

- It is a static device
- It is an electromagnetic energy conversion device

- It is a constant frequency device



So, it is having continuous magnetic circuit

So, T/F draws less magnetizing current from supply

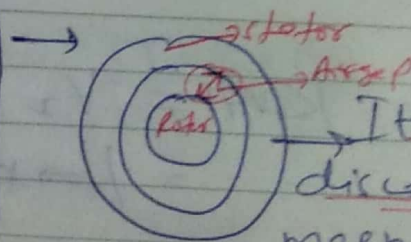
So, $I_m \rightarrow (4 + 6\% \text{ to } 7\%)$

Induction motor

- It is a mechanical device which converts electrical energy into mechanical energy

- It is an electro-mechanical energy conversion device

- It is a variable frequency device
ie $f = sf$



It is having discontinuous magnetic ckt.

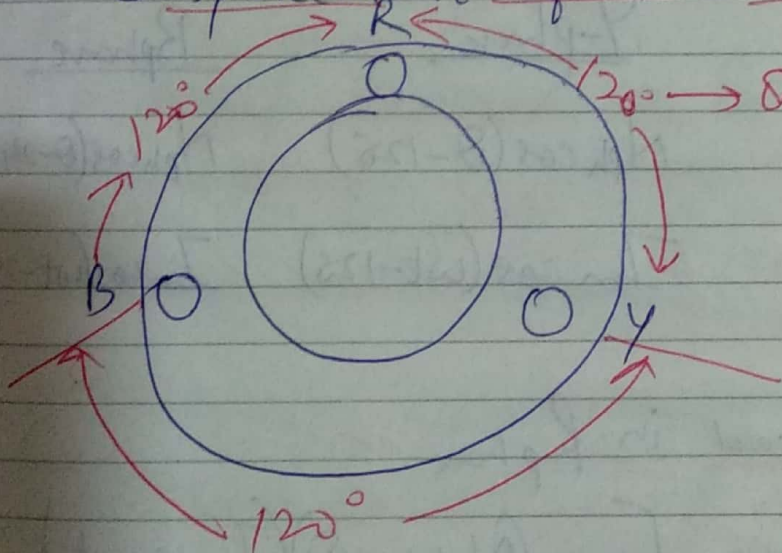
So, flux has to seek from stator to rotor by passing through air gap. Since air gap is more, reluctance is more, So, Induction motor draws more magnetizing current from supply
 $I_m = (25 \text{ to } 35\% \text{ to } 7\%)$

$\rightarrow$ Efficiency is $\uparrow$ (because of less loss)
 $\rightarrow$ Efficiency is $\downarrow$ (because of more losses)

$\Rightarrow$ Production of RMF (Rotating magnetic field)

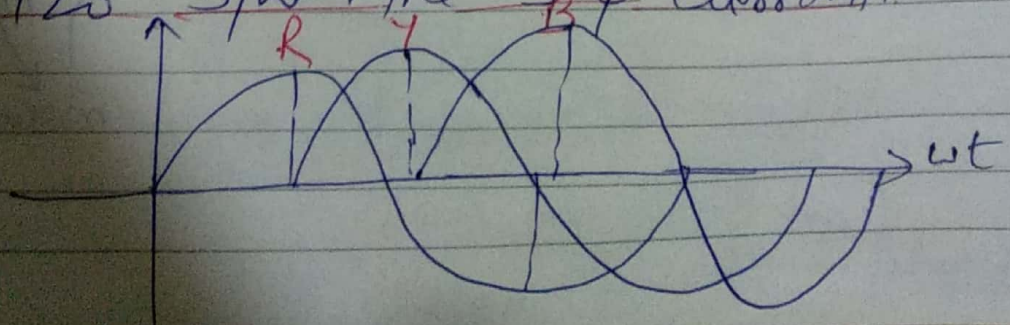
$\rightarrow$ Conditions are:-

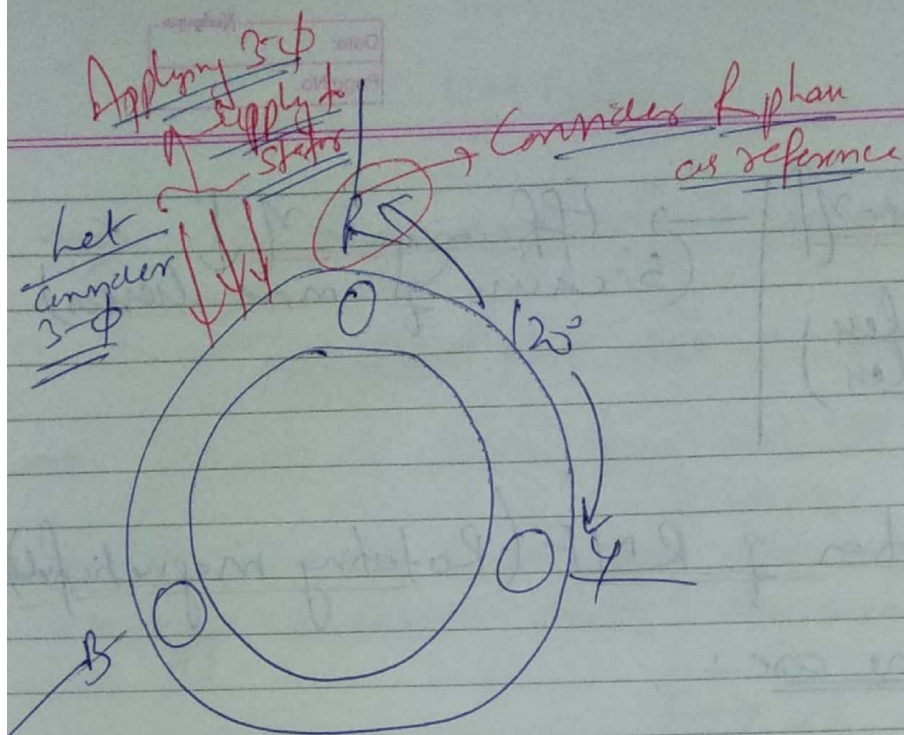
① There should be a space angle displacement of 120° b/w 3-phases



So, for any m-phase system
 $\rightarrow \frac{2\pi}{3} = 120^\circ$

② There should be a time angle displacement of 120° b/w the 3- ϕ currents





R-phase

$$N_{ph} \cos \theta$$

$$I_m \cos \omega t$$

Y-phase

$$N_{ph} \cos (\theta - 120^\circ)$$

$$I_m \cos (\omega t - 120^\circ)$$

B-phase

$$N_{ph} \cos (\theta - 240^\circ)$$

$$I_m \cos (\omega t - 240^\circ)$$

So, mmf produced in R-phase

$$\rightarrow i.e. F_R = (N_{ph} \cos \theta)(I_m \cos \omega t)$$

$$F_R = N_{ph} I_m \cos \theta \cos \omega t$$

$$\therefore F_R = F_m \cos \theta \cos \omega t$$

$$\left\{ F_m = N_{ph} I_m \right\}$$

maximum mmf

By mmf produced in Y phase

$$\rightarrow f_y = [N\phi_h \cos(\theta - 120^\circ)] [I_m \cos(\omega t - 120^\circ)]$$

$$F_y = N\phi_h I_m \cos(\theta - 120^\circ) \cos(\omega t - 120^\circ)$$

(or)

$$F_y = f_m \cos(\theta - 120^\circ) \cos(\omega t - 120^\circ)$$

By mmf produced in B phase

$$F_B = f_m \cos(\theta - 240^\circ) \cos(\omega t - 240^\circ)$$

So, The Resultant mmf in 3- ϕ m/c :-

$$\rightarrow \overline{f_{res}} = \overline{f_R} + \overline{f_y} + \overline{f_B}$$

$$\overline{f_{res}} = \frac{3}{2} f_m \cos(\theta - \omega t)$$

$$\therefore \overline{f_{res}} = f_{max} \cos(\theta - \omega t)$$

$$\text{max resultant mmf in 3-}\phi \text{ m/c} \left\{ f_{max} = \frac{3}{2} f_m \right\} \rightarrow (N\phi_h \cdot I_m)$$

2. The speed of Rotating magnetic field is always Synchronous Speed i.e.

$$N_s = \frac{120f}{P}$$

⇒ Construction of Induction m/c :-
It consists of :-

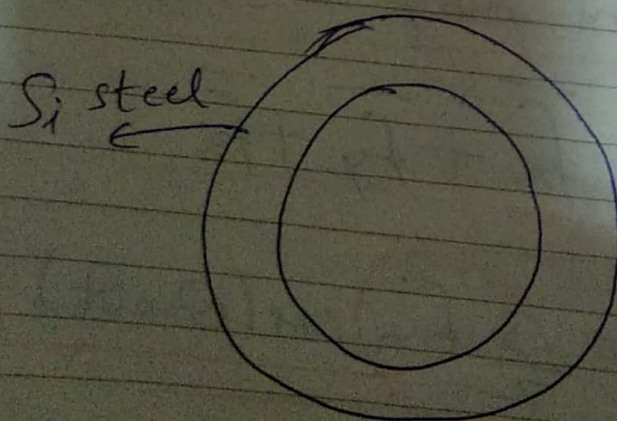
① Stator

② Rotor

Squirrel
Cage rotor

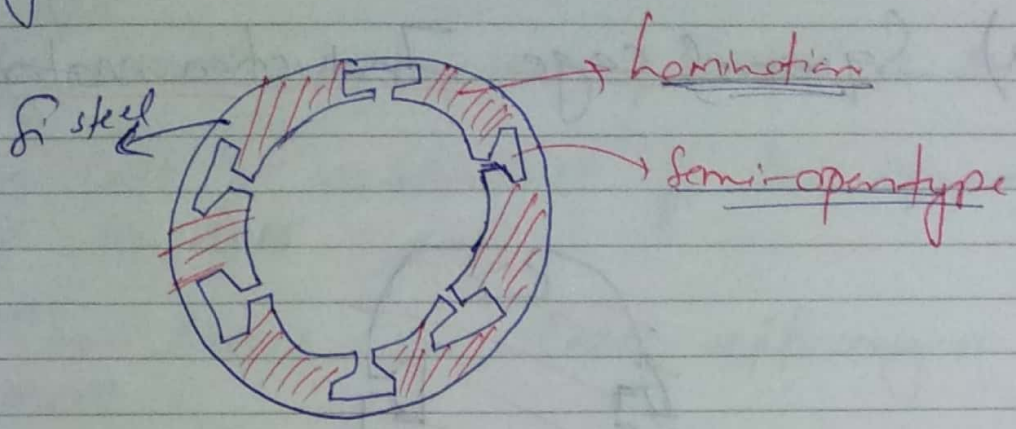
Slip ring
rotor

① Stator of Induction motor :-



Stator is made up of Silicon (Si) steel.
The slots are of semi-open type
of slots & it is laminated to reduce

eddy current losses

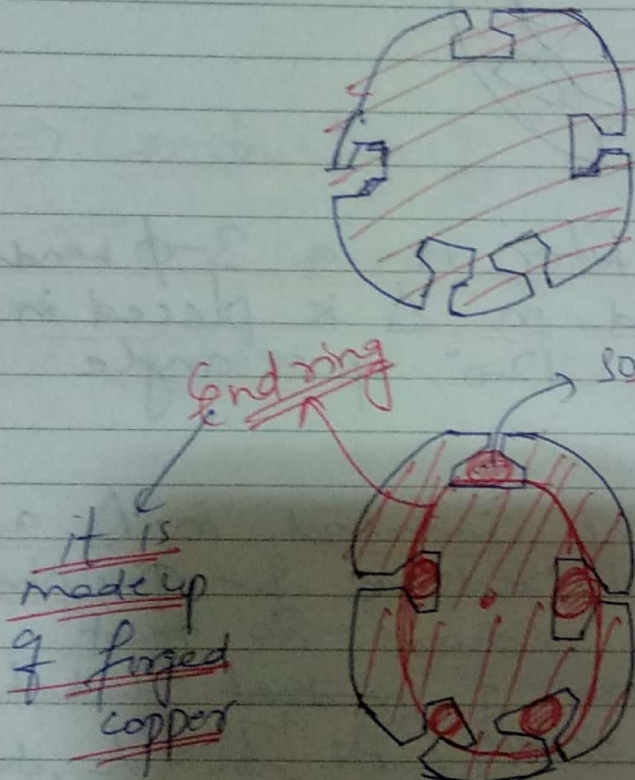


→ The purpose of stator is a 3- ϕ winding connected in Δ is placed in the slots having 120° space angle displacement

Note 3- ϕ winding connected in Δ as harmoney will reduce & the phase current is also reduces so that working operation of motor induction m/c will be better. [As $I_{ph} = \frac{I_L}{\sqrt{3}}$, so size of conductor is also smaller]

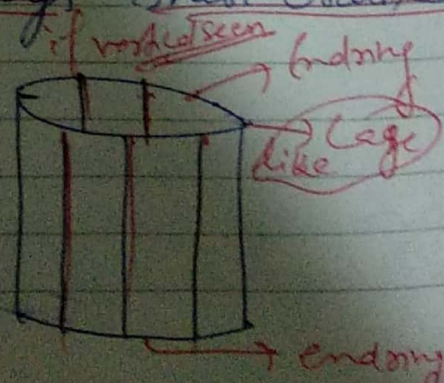
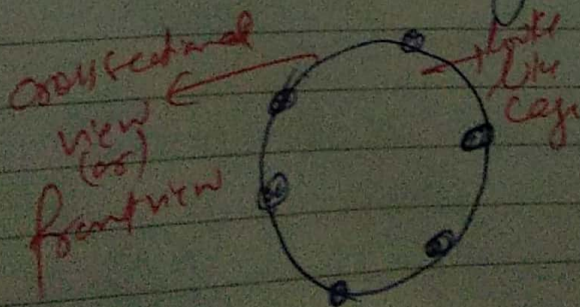
② Rotor :

a) Squirrel cage Induction motor :



In this semi-open slots heavy solid copper bars are placed & inserted & all the copper bars are short circuited with the help of end ring.

∴, In order to operate Induction m/c, rotor winding always short circuited.

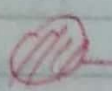


→ Advantages →

1. The winding procedure of rotor is simple.
2. The cost & maintenance of this squirrel cage is less.
3. The losses are less, efficiency is more.
4. Air gap is less.
5. It is having good running performance.

⇒ Disadvantages →

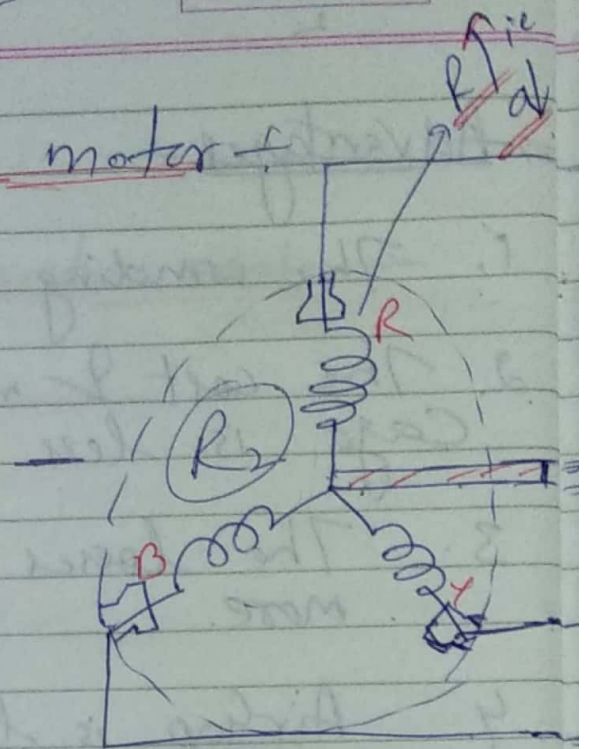
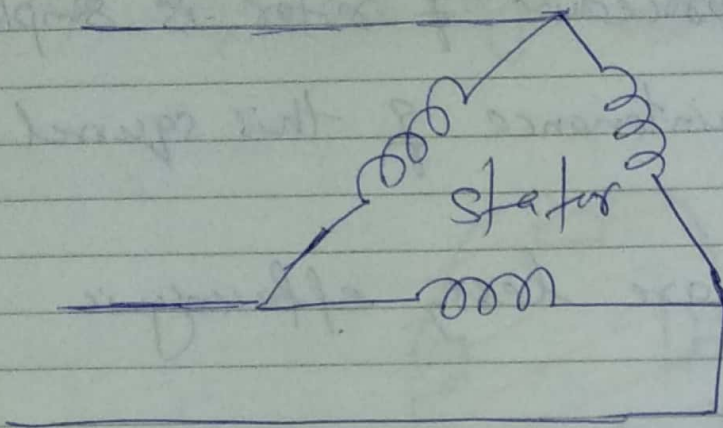
- ① As rotor resistance is less, the starting T_{st} of motor is less.

{ At  → solid copper bars so at T_{st} }
The starting current I_{st} is very high which will damage the rotor winding.

- ② The starting current I_{st} is very high which will damage the rotor winding.

- ③ Speed control methods are less.

b) Slip ring Induction motor



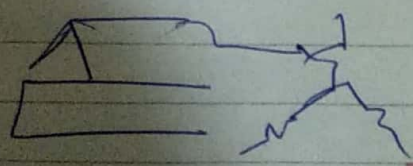
$$T_{st} \propto (R_2 + R_e)$$

→ Advantages of SRIM

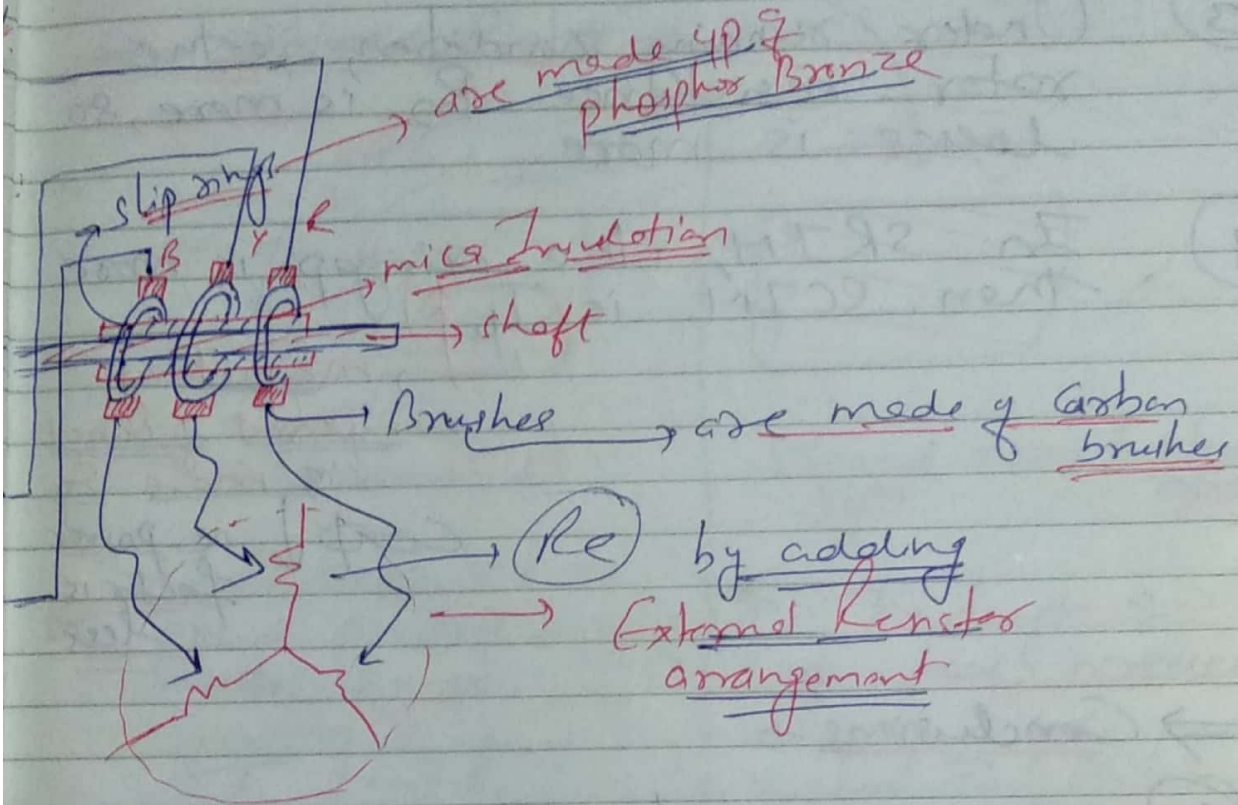
- ① As Rotor resistance is more, the T_{st} (starting Torque) is more also by adding external resistance.

Note for adding external resistor in series with each phase, the rotor winding should connect in star (Y).

if Δ connected



Not possible



② By adding external resistor the starting current drawn by Induction Motor can be limited.

③ Speed Control methods are simple (or) it is having more no. of methods.

⇒ Disadvantages -

① The size & cost of slip ring Induction motor is more

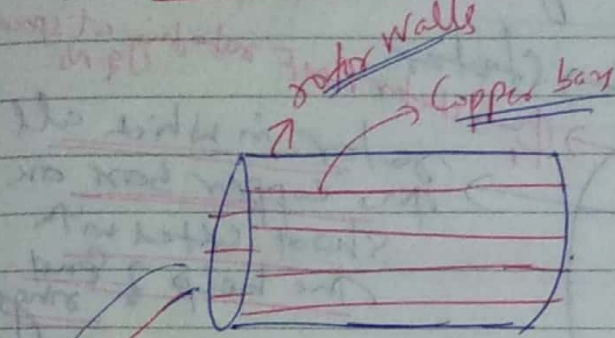
② The Maintenance is more, i.e. it is required to check regularly the brushes & slip ring arrangement.

- ③ Under running condition, as the rotor resistance R_2 is more, so losses is more
- ④ In SRIM, the airgap is more than SCIM. ie $(I_m \uparrow)$, magnetizing component of current is more so, $\cos\phi \downarrow$ ie power factor is less

⇒ Conclusions +

- ① Compared to SRIM, The SCIM has better performance under running condition. Hence it is preferred mostly in domestic purpose
- ② SCIM has a good power factor compared to SRIM.
- ③ The starting performance of SCIM can be improved by using Skewed Rotor.

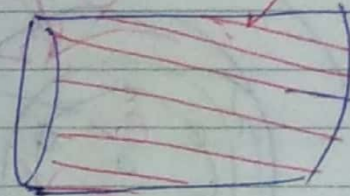
⇒ "Skewed Rotor"



⇒ Normal SCIM

Here copper bars are placed parallel to rotor walls. Hence

$$T_{st} \downarrow \rightarrow R \downarrow$$



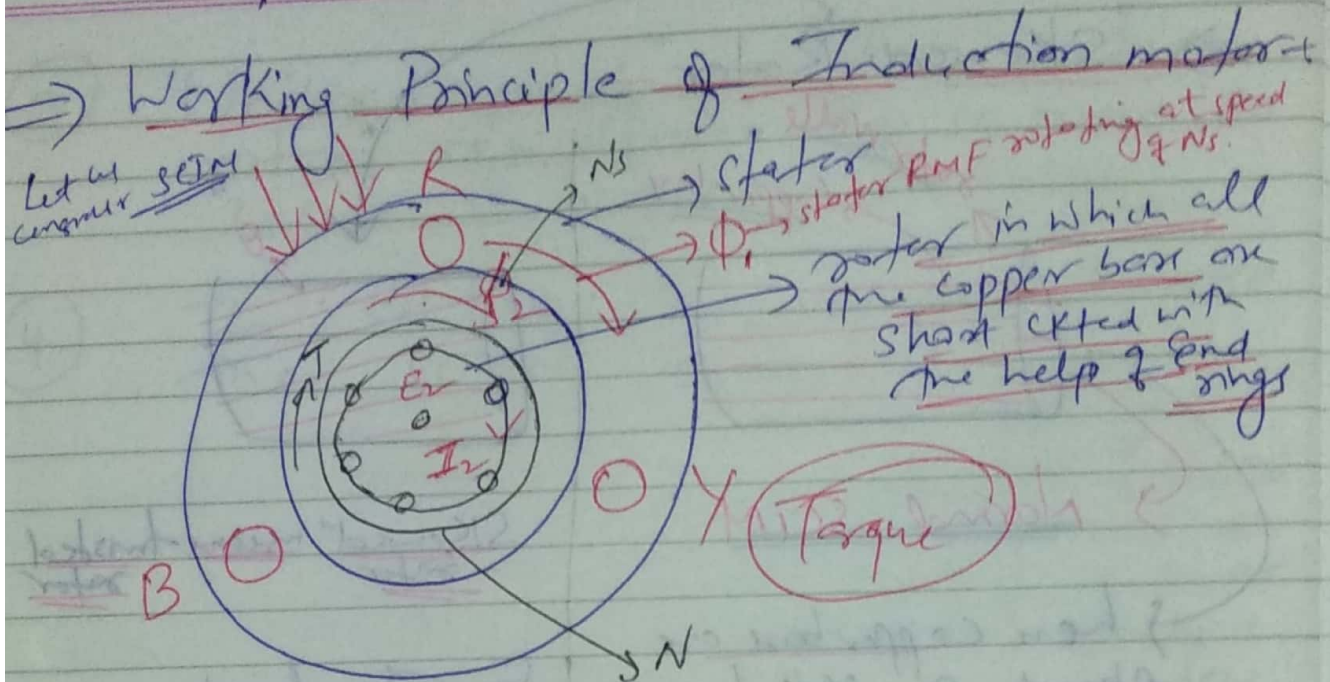
Skewed means twisted rotor

∴, the length of Copper bars increases as

$$L \uparrow \Rightarrow R \uparrow$$

$$\therefore, \text{as } R \uparrow \Rightarrow T_{st} \uparrow$$

Note Due to "skewed rotor" the starting torque improves (ie $T_{st} \uparrow$) & also harmonic reduces due to which a smooth rotation of rotor is possible.
→ means noisier rotation



So when a 3 ϕ supply is given to stator, assume an RMF (rotating magnetic field) is flowing in stator i.e. Φ_1 at synchronous speed N_s due to which at starting ~~stator~~ is rotating but rotor is stationary so due to rotating flux i.e. Φ_1 an emf will be induced in rotor i.e. E_r (as rotating flux cuts the conductor) & at the same time all rotor bars are short circuited so an current I_r will flow in rotor copper bars which produce flux again Φ_2 which should act in same direction as Φ_1 due to Lenz's law. So due to these two fluxes, i.e. stator RMF (flux) & rotor RMF (flux) a Torque will be developed i.e. torque will rotate in same direction & torque will drive the rotor in same direction so rotor starts rotating.

with Speed N . is less than N_s

$$\text{ie } N < N_s$$

So, Induction motor
also called as Asynchronous motor

Note When a 3- ϕ supply is given to stator, an RMF (ϕ_1) will produce rotating at N_s . Due to ϕ_1 an emf will induce in the rotor winding which causes a rotor flux ϕ_2 .

To satisfy Lenz's law, the rotor flux ϕ_2 will also rotate in the same direction of ϕ_1 at N_s .

During this a Torque will develop which draws the rotor in the same direction. But the rotor rotates at Speed N always which is less than N_s .

$$\text{ie } N < N_s$$

The relative speed b/w the rmf & rotor is $N_s - N$ which is called slip speed

$$\therefore \text{slip speed of Induction motor} = N_s - N$$

$$\text{In p.u. slip, } s = \frac{\text{slip speed}}{N_s} = \frac{N_s - N}{N_s}$$

per unit

Name - Prashant Nigam
 Sub - EM
 ↳ (Electrical machines)

$$s = \frac{N_s - N}{N_s}$$

↳ slip of Induction motor

(or)

$$s N_s = N_s - N$$

$$\rightarrow N = N_s (1 - s) \quad (\text{rpm})$$

↳ Speed of motor in rpm

Ans

So, speed N depends upon two parameters

- 1) N_s → Synchronous speed &
- 2) s → slip

→ At starting

Let stator freq = f

∴ Rotor freq ' f_2 ' = ?

As $f_2 = \frac{(N_s)P}{120} = f$

$f_2 = f$ → at starting condition

But Under Running Condⁿ —
 (Note) N_s is rotating at speed N_s
 Let rotor is rotating at speed N

$f_{2r} = \frac{(N_s - N)P}{120} \propto \frac{N_s}{N_s}$

$\Rightarrow = \frac{N_s - N}{N_s} \times \frac{N_s P}{120}$

$f_{2r} \Rightarrow s f$ → Under running Condⁿ

↓
slip frequency of induction motor

Note

1. The speed of Rotor w.r.t stator is N_s

2. The speed of Rotor w.r.t stator is N

3. The speed of Rotor w.r.t RMF is $(N - N_s)$

4. The speed of stator m.f with respect to rotor m.f is $N_s - N$

⇒ Changes in Rotor parameters :-

⇒ Rotor parameters :- it will remain same

1. $R_2 \rightarrow$ Rotor Resistance

2. $X_2 \rightarrow$ Rotor Reactance

So, $Z_2 = (R_2 + jX_2) \rightarrow$ Rotor Impedance

3. $E_2 \rightarrow$ Rotor EMF

4. $I_2 \rightarrow$ Rotor Current

At Starting

$$f_2 = f$$

$$E_{2ph} = 4.44 B_m A_n f N_{2ph} K_w r_2$$

winding factor K_w refer

$$Z_2 = R_2 + jX_2$$

$$X_2 = 2\pi fL$$

$$|Z_2| = \sqrt{R_2^2 + X_2^2}$$

$$I_2 = \frac{E_2}{\sqrt{R_2^2 + X_2^2}}$$

rotor current at standstill

Now Under Running Condⁿ :-

$$f_{2r} = sf$$

$$E_{2r} = 4.44 B_m A_n (sf) N_{2ph} K_w r_2$$

(or)

$$E_{2r} = s[E_{2ph}]$$

$$X_{2r} = 2\pi (sf) \cdot L = sX_2$$

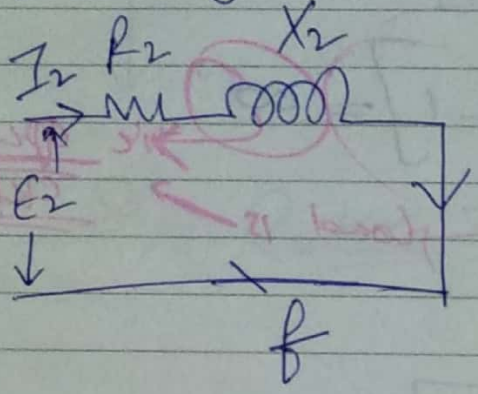
$$Z_2 = \sqrt{R_2^2 + (sX_2)^2}$$

$$I_{2r} = \frac{E_{2r}}{\sqrt{R_2^2 + (sX_2)^2}}$$

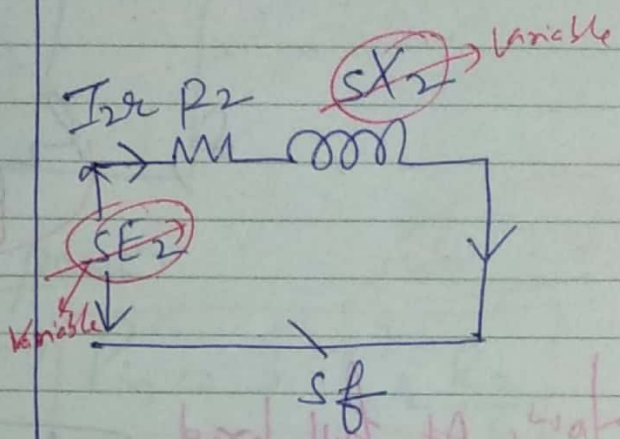
rotor current at running condⁿ
 $= \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}}$

→ Rotor Equivalent Ckt :-

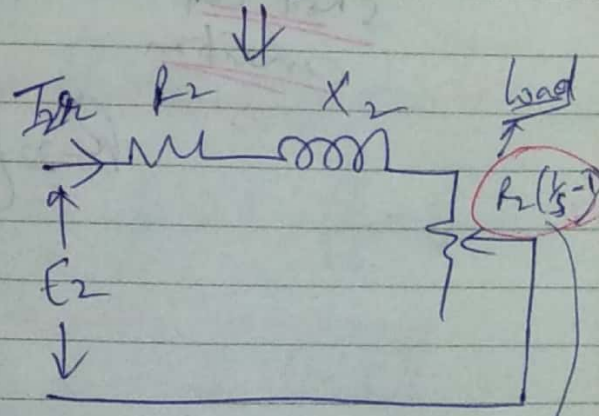
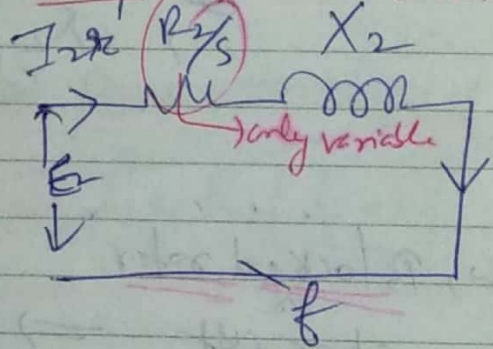
At starting



At Running



So, Dividing all the parameters of rotor with slip ckt becomes



So, $R_2 \left[\frac{1-s}{s} \right] \rightarrow$ mechanical load

So, At No-load $\Rightarrow \boxed{N \approx N_s}$

hence $s = \frac{N_s - N}{N_s} \Rightarrow \boxed{s=0}$

So, $R_2 \left[\frac{1}{s} - 1 \right] = \infty$ gives open
circuit
load is

Now, At full load

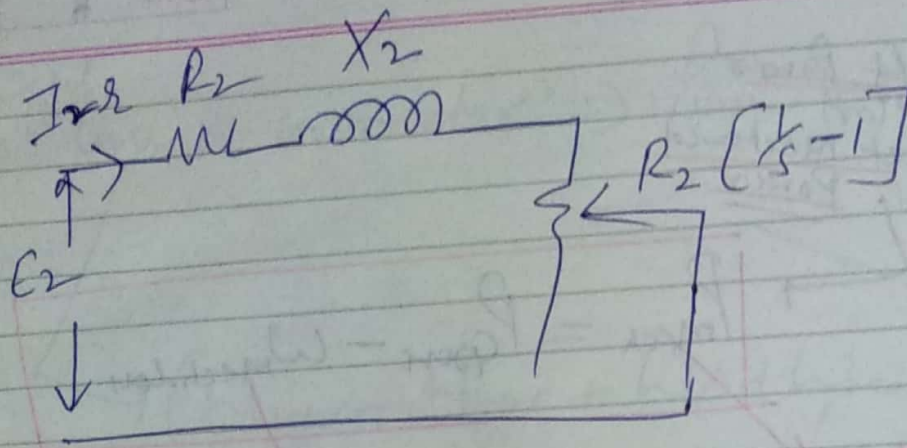
$\boxed{N \neq N_s} \xrightarrow{\text{then}} s = 0.04, 0.06$
slip is present

So, $R_2 \left(\frac{1}{s} - 1 \right) \rightarrow$ high resistance

Now, Blocked rotor
short circuit
condition

$\Rightarrow \boxed{N=0} \xrightarrow{\text{then}} \boxed{s=1}$

$R_2 \left[\frac{1}{s} - 1 \right] \Rightarrow 0$, which
indicates short
circuit



So, Rotor I/p power

(for 3-φ)

$$P_{in} = 3 I_{2r}^2 \frac{R_2}{s}$$

Rotor Cu loss $\Rightarrow P_{cu} = 3 I_{2r}^2 R_2$

& The gross mechanical power developed

$$P_{gross} = 3 I_{2r}^2 R_2 \left[\frac{1}{s} - 1 \right]$$

→ Relation b/w Rotor Power Equations

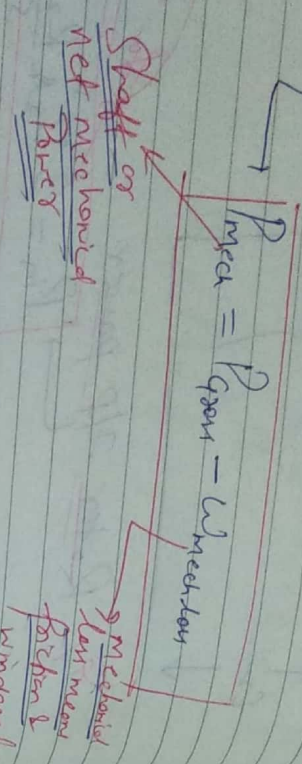
$$P_{in} : P_{cu} : P_{gross} = \frac{1}{s} : 1 : \left(\frac{1}{s} - 1 \right)$$

$$P_{in} : P_{cu} : P_{gross} = 1 : s : (1-s)$$

So, if we want

$$\frac{P_{in}}{P_{cu}} = \frac{1}{s} \quad \bigg/ \quad \frac{P_m}{P_{gross}} = \left(\frac{1}{1-s} \right) \bigg/ \frac{P_{cu}}{P_{gross}} = \frac{s}{1-s}$$

Simultaneous
ask shaft Power
net mechanical
Power



Q A 3- ϕ , 40kW, SCIM has an efficiency of 90%. The stator & rotor copper loss each equally to iron loss & mechanical loss is 1/3rd of from loss. Then find the (3) i) Stator Cu loss

Soln Given $\eta = 90\%$

$$\eta = \frac{P_{out}}{P_{in}} = \frac{P_{mech}}{P_{in}}$$

$$90\% = 0.9 = \frac{40kW}{P_{in}}$$

$$P_{in} = \frac{40}{0.9} = 44.44 kW$$

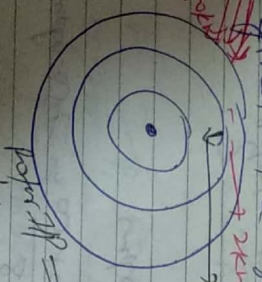
Now gives in motor $\Rightarrow$ Input - net

$$P_{in} - P_{loss} = P_{out}$$

$$W_i + W_r + W_{sc} + W_{rc} = P_{in} - P_{out}$$

$$W_i = \frac{1}{3} (P_{in} - P_{out})$$

Q A 3- ϕ , 40kW, 50Hz induction motor draws 50kW from supply. The rotor Cu loss are 2kW. Find the stator losses are 2kW. Then find the rotor Cu loss.



$$P_{in} = 50kW$$

$$P_{sc} = 2kW$$

$$P_{rc} = 2kW$$

$$\Rightarrow \frac{P_m}{P_{cu}} = \frac{1}{5}$$

So, we have obtained $\frac{P_m}{P_{cu}} = \frac{1}{5}$

$$\Rightarrow P_{cu} = P_m \times 5$$

$$\Rightarrow (48 \text{ kW}) \times \frac{1}{5}$$

$$P_{cu} = 1.6 \text{ kW}$$

Plata Cu has

An 8-pole, 3-phase 50 Hz induction motor is operating at a speed of 700 rpm. The frequency of the rotor current is _____?

Solⁿ Given $P = 8$
 $N = 700 \text{ rpm}$

$$\therefore N_s = \frac{120 \times 50}{8} = 750 \text{ rpm}$$

$$\text{hence } s = \frac{N_s - N}{N_s} = \frac{750 - 700}{750} = 0.067$$

$$s = 0.067$$

$$f_r = sf$$

$$f_r = 0.067 \times 50$$

$$f_r = 3.33 \text{ Hz}$$

freq of rotor current

The slip of induction motor normally depends on _____

- a) Rotor speed b) Synchronous speed
- c) Shaft Torque d) Core loss component

Solⁿ $s = \frac{N_s - N}{N_s}$ (or) $s = N_s(1-s)$

A 3-phase, 50 Hz slip ring induction motor has 6 poles on stator & 4 poles on rotor, then the possible speed of motor is _____

- a) 960 rpm b) 1440 rpm c) 1400 rpm

d) 0 rpm

Note for different no. of poles on stator & rotor, the induction motor operation is not possible.

Q The Power I/P to a 3- ϕ 7 pole induction motor is 70 kW. The stator Cu loss & iron loss are 2.5 kW. Find the mechanical Power if the motor is running with a slip of 5%.

- Solⁿ
- a) 67.5 kW b) 64.125 kW
c) 65 kW d) 62.5 kW

Given Stator I/P (P_{in}) = 70 kW

Stator loss $\Rightarrow$ 2.5 kW

$$s = 5\% = 0.05$$

$$P_{gross} = ?$$

We know

$$\frac{P_{in}}{P_{gross}} = \frac{1}{1-s}$$

So Stator O/P $\Rightarrow P_m = \frac{P_{in} - \text{Stator loss}}{1-s}$
 $\Rightarrow 70 - 2.5 = 67.5 \text{ kW}$

So, $P_{gross} = P_m [1-s] = 67.5 [1-0.05]$

Ans $P_{gross} = 64.125 \text{ kW}$

Q The rotor I/P of a 3- ϕ , 6 pole 50 Hz induction motor alternates at 3 Hz. Then the speed of motor is _____ rpm?

Solⁿ Given $p=6$, $f=50 \text{ Hz}$

$$N_s = \frac{120f}{p} = \frac{120 \times 50}{6}$$

$$N_s = 1000 \text{ rpm}$$

Rotor I/P

$$\Rightarrow 342 = sf$$

$$\Rightarrow s = \frac{3}{50} = 0.06$$

$$s = 0.06$$

Speed of motor

$$N = N_s (1-s)$$

$$N = 940 \text{ rpm}$$

- 8 The Power IP to 415V, at 975 rpm is 40kW. The losses are 1kW & mechanical & motor is 90% efficiency
- a) 92.5% b) 90% c) 91% d) 87%

8.12 Given

Stator IP $\Rightarrow$ 40kW

$N = 975$ rpm

$\eta = ?$

At $P_m = P_{net}$ $\rightarrow$ Gross - loss

Stator O/P $\Rightarrow$ Rotor I/P (P_m) $\Rightarrow$ 40kW

rotor I/P (P_m) = 39kW

As $\frac{P_m}{P_{gross}} = \frac{1}{1.5}$ (1)

$\rightarrow S = \frac{N_s - N}{N_s} = \frac{1000 - 975}{1000} = 0.025$

$S = 0.025$

8 ep (1) because

$\rightarrow P_{gross} = P_m (1.5)$

$= 39 [1 + 0.025]$

$P_{gross} = 39.025 \text{ kW}$

$P_{net} = P_{gross} - \text{loss}$

$= 38.025 - 2$

$P_{net} = 36.025 \text{ kW}$

Hence $\eta = \frac{36.025}{40} = 90\%$

$\eta = 90\%$

⇒ Torque Eqn of Induction motor :-

Due to interaction b/w
stator and ϕ_1 & rotor ϕ_2

⇒ $T \propto P_{\text{gross}}$

⇒ $T = \frac{P_{\text{gross}}}{\omega_s} = \frac{3 I_2^2 R_2 (s-1)}{2\pi N_s / 60}$

$= \frac{180}{2\pi N_s} \frac{I_2^2}{R_2} [s-1]$

$\frac{180}{2\pi N_s} \frac{I_2^2}{R_2} [s-1] = \frac{180}{2\pi N_s} \left[\frac{s \cdot E_2^2}{(R_2^2 + (sX_2)^2)} \right] \frac{R_2}{s} [1-s]$

$T = \frac{180}{2\pi N_s} \left[\frac{s \cdot E_2^2}{(R_2^2 + (sX_2)^2)} \right] \cdot R_2 \left[\frac{N_s}{N_r} \right]$

$T = \frac{180}{2\pi N_s} \left[\frac{s \cdot E_2^2 \cdot R_2}{R_2^2 + (sX_2)^2} \right] \frac{N_s}{N_r}$

⇒ Generalised eqn of Torque

Note from op (1) $T \propto E_2^2 \propto V_1^2$

If the supply voltage V_1 is reduced by 10%,
then $T \propto (0.9V_1)^2 \propto 0.81V_1^2$

$T \downarrow$ by 19%

2. $T \propto s \Rightarrow$ if slip is low (ie 0.01, 0.02, 0.03, 0.04)

At $S = \frac{N_s - N}{N_s}$

if $N = N_s$, then $S = 0$

if $N = 0$, then $S = 1$
means just starting

$T \propto 1/s \Rightarrow$ if slip is high (ie $S = 1$)
(ie 0.09, 0.08, 0.07, 0.06)

⇒ Maximum torque = $(T_{max})_s$

$$T = \frac{180}{2\pi N_s} \left[\frac{S E_2^2 R_2}{R_2^2 + (SX_2)^2} \right]$$

also called Pull out torque (or) Breakdown torque (or) Stalling torque.

It is defined as the maximum torque given by the induction motor at the edge it becomes unstable i.e. Pulling out the induction motor from one region to other region.

⇒ Condition for maximum torque:

$$\frac{dT}{dS} = 0$$

So, we get

$$SX_2 = R_2$$

hence

$$S = \frac{R_2}{X_2}$$

maximum slip corresponds to max. torque

Maximum torque is

$$T_{max} = \frac{180}{2\pi N_s} \left[\frac{S_m E_2^2 R_2}{R_2^2 + (S_m X_2)^2} \right]$$

$$T_{max} \rightarrow \frac{180}{2\pi N_s} \cdot \frac{R_2^2 X_2}{R_2^2 + \left(\frac{R_2}{X_2} \cdot X_2 \right)^2}$$

$$= \frac{180}{2\pi N_s} \cdot \frac{E_2^2 R_2}{2R_2 X_2}$$

$$T_{max} = \frac{180}{2\pi N_s} \cdot \frac{E_2^2}{2X_2}$$

max. torque produced in induction motor

Note

$$T_{max} \propto E_2^2 \propto V^2$$

$$T_{max} \propto \frac{1}{X_2}$$

$X_2 \propto$ motor reactance depends on frequency

$$(T_{max})_{SCIM} > (T_{max})_{SPIM}$$

⇒ Starting torque of induction motor

At starting

$N=0$ & $s=1$

$$T_{st} = \frac{180}{2\pi N_s} \left[\frac{1 \cdot E_2^2 \cdot R_2}{R_2^2 + X_2^2} \right]$$

$$T_{st} = \frac{180}{2\pi N_s} \left[\frac{E_2^2 R_2}{R_2^2 + X_2^2} \right]$$

hence $T_{st} \propto R_2$

only possible in
slip ring induction
motor i.e.
 $T_{st} \propto (R_2 + R_e)$

Note
1. $T_{st} \propto E_2^2 \propto V_1^2$

2. $T_{st} \propto \left(\frac{1}{R_2^2 + X_2^2} \right)$

3. $T_{st} \propto \frac{1}{X_2}$

⇒ Condition for maximum starting torque

At for max. torque

$s_{m} = \frac{R_2}{X_2}$ (1)

for starting
max. torque $s=1$

So eq (1) become
 $\Rightarrow 1 = \frac{R_2}{X_2} \Rightarrow R_2 = X_2$

Note
1. $T_{st} = \frac{180}{2\pi N_s} \cdot \frac{s E_2^2 R_2}{R_2^2 + (s X_2)^2}$

2. $T_{max} = \frac{180}{2\pi N_s} \cdot \frac{E_2^2}{2 X_2}$

3. $T_{st} = \frac{180}{2\pi N_s} \cdot \frac{E_2^2 R_2}{R_2^2 + X_2^2}$

$$\frac{T_{st}}{T_{max}} = \frac{2 \cdot S_m \cdot S_{pe}}{S_m^2 + S_{pe}^2}$$

decide the stability of induction motor

$$\frac{T_{st}}{T_{max}} = \frac{2 \cdot S_m}{1 + S_m^2}$$

$$S_m = \frac{R_2}{X_2}$$

max. torque is taken as reference:

Q 3- ϕ Induction motor has a full load slip of 80% of S_m , then the T_{max} term of T_{max} is _____

Soln Given $S_{pe} = 80\%$ of S_m

we know

$$\frac{T_{pe}}{T_{max}} = \frac{2 \cdot S_m \cdot S_{pe}}{S_m^2 + S_{pe}^2}$$

(or)

$$\begin{aligned} \rightarrow \frac{T_{pe}}{T_{max}} &= \frac{2}{\frac{S_m}{S_{pe}} + \frac{S_{pe}}{S_m}} \\ &= \frac{2}{\frac{1}{0.8} + \frac{0.8}{1}} = 0.98 \end{aligned}$$

Q In a 3- ϕ Induction motor, the T_{st} developed is 120% of T_{max} developed is 200% of rated torque. Then slip corresponds to maximum torque is _____

- a) 0.33 b) 3 c) 3.25 d) 0.25

Soln we know $S_m = ?$

Given $T_{st} = 120\%$ $\Rightarrow 1.2 T_{max}$ Let rated torque
Q $T_{max} = 200\%$ $= 2T$

Soln we know

$$\frac{T_{st}}{T_{max}} = \frac{2 \cdot S_m}{1 + S_m^2}$$

$$\Rightarrow \frac{1.2T}{2T} = \frac{2 \cdot S_m}{1 + S_m^2}$$

$$\Rightarrow 1.2 S_m^2 + 1.2 = 4 S_m$$

$$\Rightarrow 1.2 S_m^2 + 1.2 - 4 S_m = 0$$

So after solving we get

$$S_m = 0.33 \text{ and } 3x$$

So $S_m = 0.33$ Ans

Q. A 3- ϕ , 50 Hz, 6 pole induction motor has a rotor resistance of 0.1 Ω & reactance of 0.92 Ω . Neglect the voltage drop in stator & assume that the rotor resistance is constant. Given $s_{pl} = 3\%$, the ratio of max Torque to full load Torque is _____?

- Soln a) 1.527 b) 1.712 c) 1.94 d) 2.134

Ans
$$\frac{T_{pe}}{T_{max}} = \frac{2 s_m s_{pl}}{s_m^2 + s_{pl}^2}$$

$$\frac{T_{max}}{T_{pl}} = ?$$

$$\frac{T_{max}}{T_{pe}} = \frac{s_m^2 + s_{pl}^2}{2 s_m s_{pl}} \quad (1)$$

Given $R_2 = 0.1 \Omega$ & $X_2 = 0.92 \Omega$

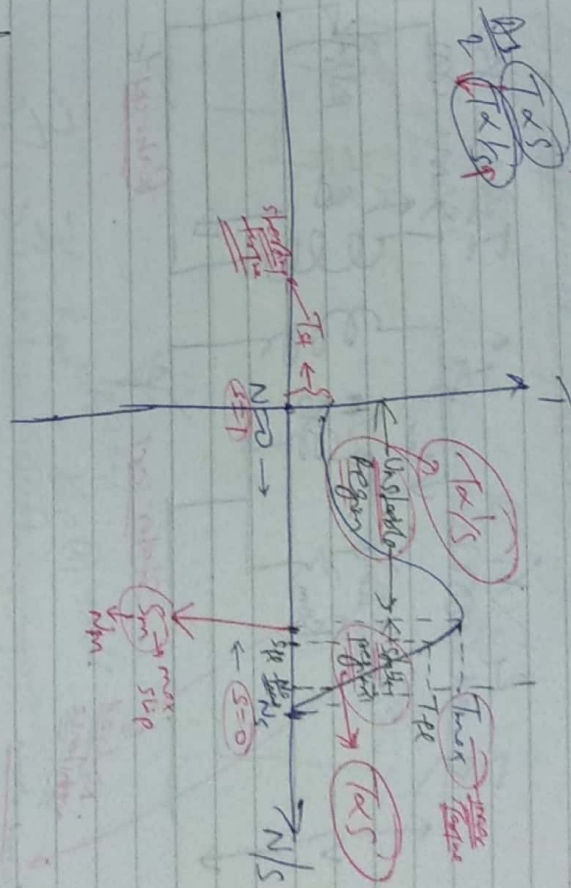
$$s_1, s_m = \frac{R_2}{X_2} = \frac{0.1}{0.92}$$

$$s_m = 0.108$$

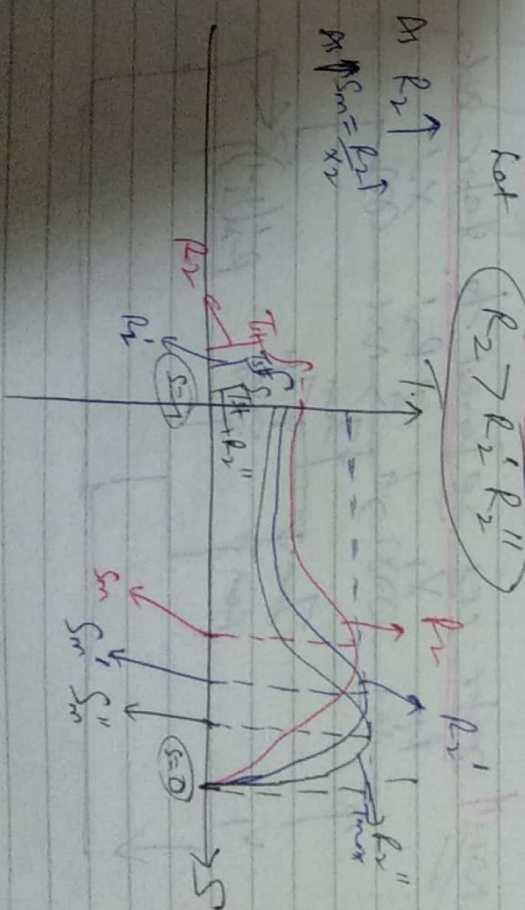
eq (1) becomes
$$\frac{T_{max}}{T_{pl}} = \frac{(0.108)^2 + (0.03)^2}{2 \times 0.108 \times 0.03}$$

$$= 1.938 \approx 1.94$$

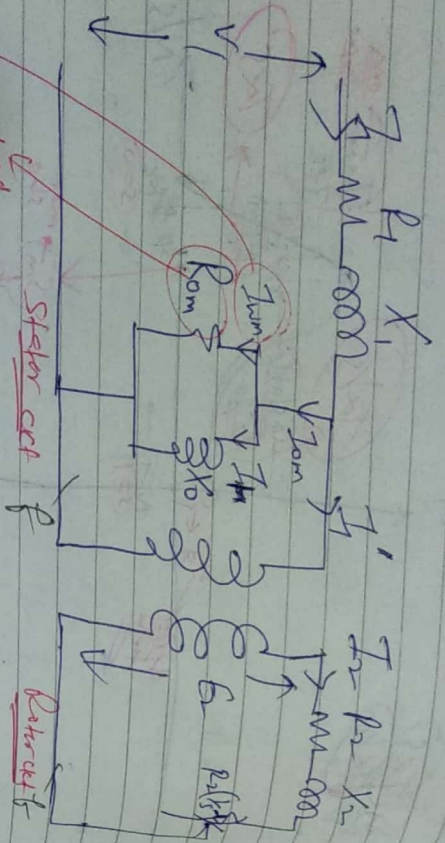
$\Rightarrow$ Torque-Slip characteristics is a curve of speed



Now Torque-slip characteristics for different rotor reactances



⇒ Equivalent circuit of induction motor



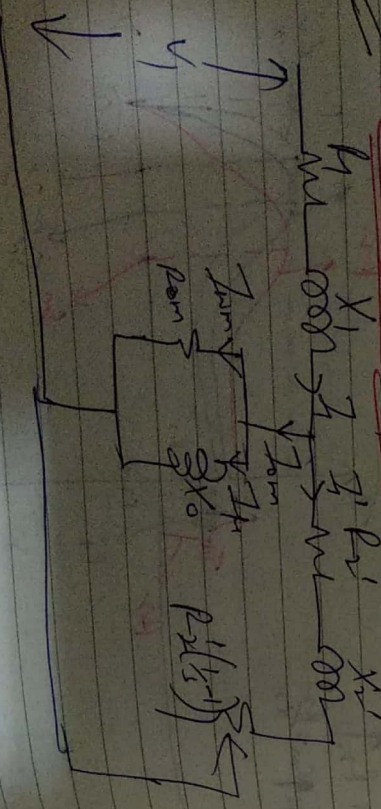
No load
resistance

Stator
circuit

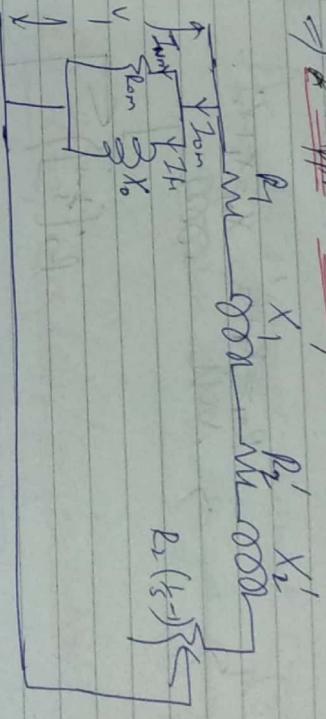
Rotator
circuit

Current drawn
by induction motor
in order to meet
Rational & mechanical loss.

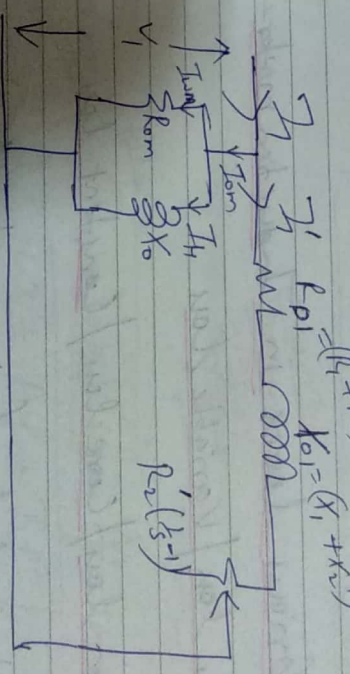
Now for ckt referred to stator side.



∴ approximate equivalent ckt is

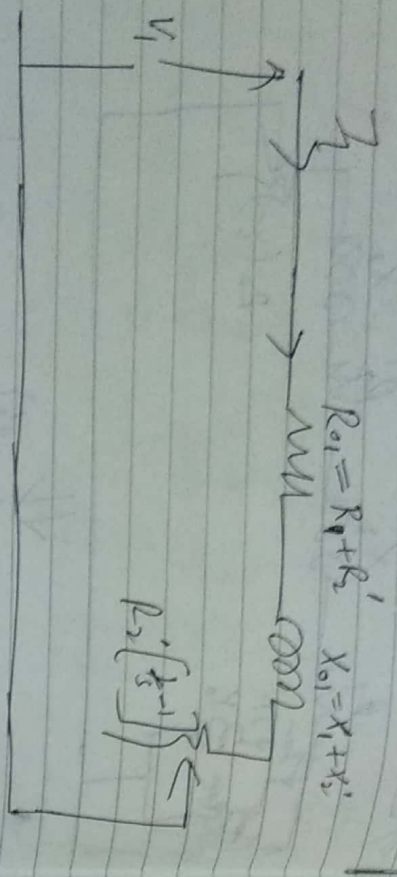


$R_{01} = (R_1 + R_2)$
 $X_{01} = (X_1 + X_2)$



inductance inductance

⇒ 2nd approximate equivalent circuit:



⇒ Various losses in induction motor:

1. Cu loss / Variable loss
2. Iron loss / Core loss / Constant loss
3. Mechanical loss (also constant loss)

① Cu loss / Variable loss:

$$P_{Cu} = I_1^2 R_1 + I_2^2 R_2$$

$$P_{Cu} = I_1^2 R_1 + I_2^2 R_2$$

$$= I_1^2 [R_1] \text{ or } I_2^2 [R_2]$$

∴ Cu loss ∝ I² ∝ load

As load change, current changes
∴ Cu loss also get change

② Iron / Core loss / Constant loss

$$W_i = (k_1 + k_2 \omega^2)$$

due to pressure & alternating flux

when $B_m = \text{constant} \Rightarrow \phi = \text{constant}$

$$P_i = [W_i = A \phi + B \phi^2]$$

when $B_m \neq \text{constant} \Rightarrow \phi \neq \text{constant}$

$$P_i = [W_i = A V_1^{1.6} + B V_1^2]$$

5/2/25

2. Running $\Rightarrow$ stater $\rightarrow$ stater

56 very low
5242

Due to this, the iron loss in induction motor is due to stator only. Since a rotor loss is low (negligible), thus the iron loss is also negligible.

③ Mechanical loss

91
fashion losses

⑤ Winding losses

due to $\rightarrow$ Rough pattern lay
 $\downarrow$ $\rightarrow$ Bearing fashion lay

Beating fiction law

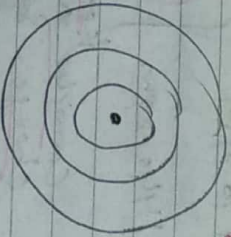
Present ^m both
10 Sept 8 5.14

① Fruch frühen das

→ It is the friction takes place b/w
The stationary brakes & rotating sliprings
So, it can be reduced by using the

→ Self dependent maternal

⑨ Beany frictionless $\rightarrow$ we could lubricate $\rightarrow$ to reduce this loss



Outlet

Uebn'

Bernoulli's eqn (etc)

K

Fractal

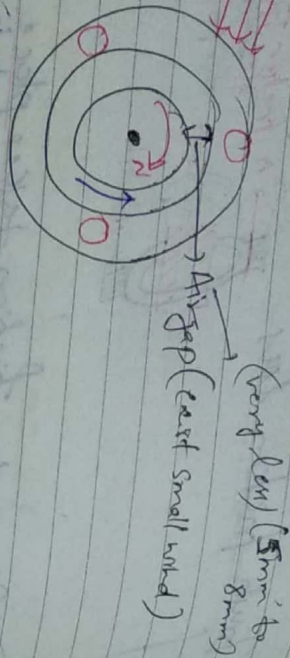
Connected to speaker

after (small) pollen → for birds including
type like major

(Steel
2011/12)

(steel ball)

6) Windage loss - it is negligible loss in induction motor



During the rotation of rotor, the wind present in the air gap will oppose the direction of rotation which is called Windage loss.

Note 8, friction & Windage loss depends upon Speed of Rotor.

Lot No load $\rightarrow N = 1490 \text{ rpm}$

But full load $\rightarrow N = 1440 \text{ rpm}$

80% change in speed will be 100% from it is consider as negligible change, so induction motor considered as constant m/c motor.

Note The change in speed of induction motor from no-load to full load is negligible. Hence it is treated as constant speed motor.

$\therefore$ The mechanical loss which depends on speed is also treated as a constant loss.

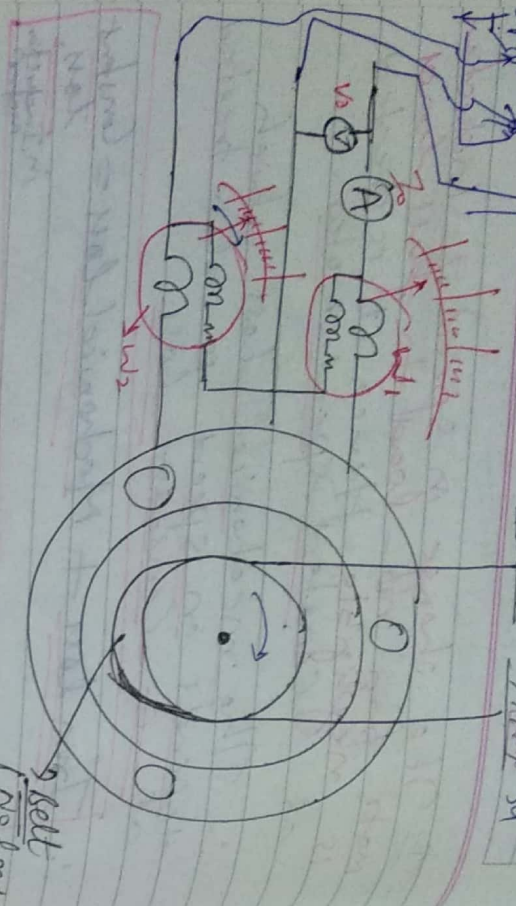
$\therefore$ Iron loss + Mechanical loss = constant loss in induction motor

$\Rightarrow$ Tests on Induction Motor:-

1. No load test
2. Blocked Rotor test

① No load test - its purpose is to find constant lower

to find no-load branch parameter ie [Rm & Xo]



Procedure

1. Connect the circuit as shown
2. observe the induction meter should be under no-load
3. Apply rated voltage & rated frequency vary ϕ & V
4. Take the readings of Ammeter (A), Wattmeter (W), & Voltmeter (V)

Let

V_0	I_0	W_1	W_2	W_0
		+ve	-ve	$(W_1 - W_2)$

Power consumed under no load

Note: No load pf is $\cos \phi$ of induction motor under no load is 0.45 to 0.70

So when $\cos \phi < 0.5$ from $W_1 = +ve$ & $W_2 = -ve$ deflection is in opposite direction

Then we cannot read any value for interchange and the difference of current will be the same.

a) To find $\cos \phi$

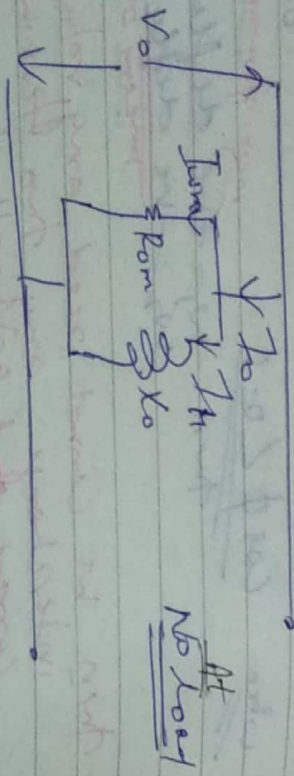
$$W_0 = (W_1 - W_2) = \text{Constant loss} + \text{a small loss}$$

Constant loss $\approx W_0$

(Since by neglecting small W_0 loss)

5) To find No load branch parameters R_{om} & X_o

8, consider it approximate as



$$I_{om} = \frac{V_{oph}}{R_{om}} \Rightarrow R_{om} = \frac{V_{oph}}{I_{om}}$$

$$X_o = \frac{V_{oph}}{I_m}$$

$$I_{wm} = I_{om} \cos \phi_o$$

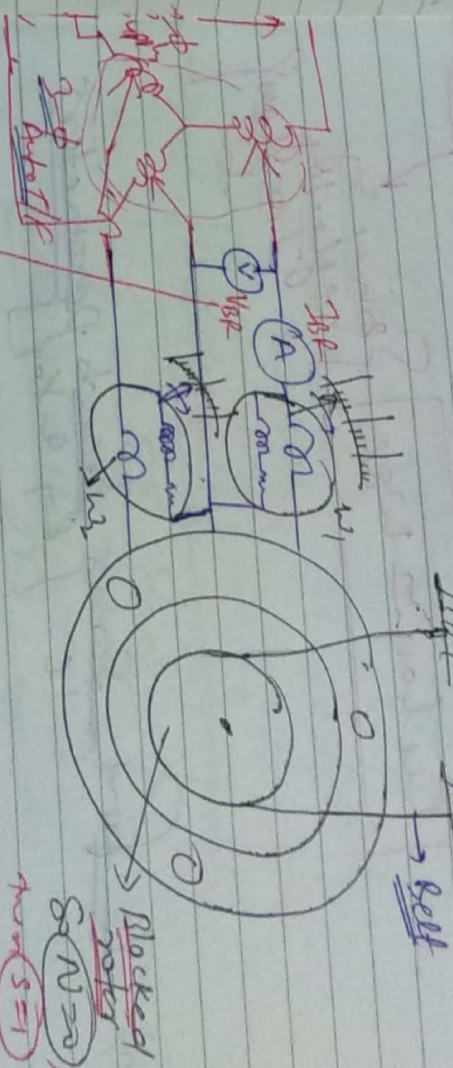
$$I_m = I_{om} \sin \phi_o$$

$$\cos \phi_o = \frac{W_o}{3 V_{ph} I_{om}}$$

A I_{om} is line current. So convert it to phase current as $\frac{I_{om}}{\sqrt{3}}$

should conduct test at Rated current

9) Blocked Rotor test -
Purpose is to find the Cu losses & Variable losses & to find the series branch parameters R_{e1} & X_{e1}



8 to 12% of V_{rated} (440V)

Under this pf $\cos \phi_{br} = 0.8$

both wattmeter ready (are) short deflection

V_{br}	I_{br}	W_1	W_2	$W_{3\phi}$
				Total power consumed under blocked rotor

a) To find $i_{u, \text{loss}}$

$\Rightarrow \omega_{BR} = (\omega_1 + \omega_2) = i_{u, \text{loss}} +$

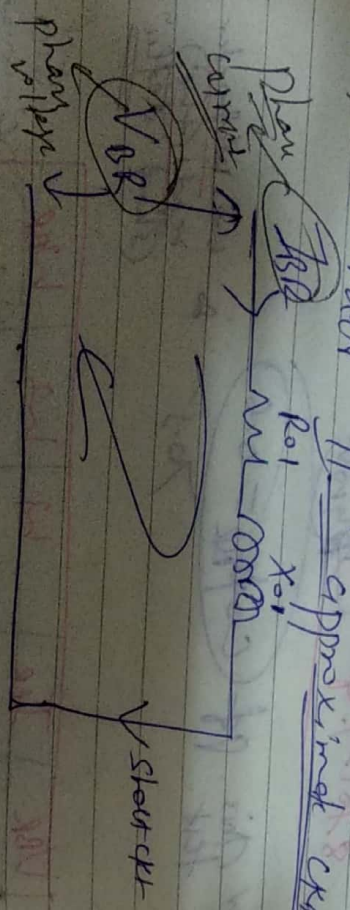
a small amount of $i_{u, \text{loss}}$

$\Rightarrow i_{u, \text{loss}} \approx \omega_{BR}$

Since by neglecting $i_{u, \text{loss}}$

b) To find series branch parameters i.e. $[R_0, X_0]$

So, consider the approximate ckt.



So, $V_{BR} = I_{BR} [R_0 + jX_0]$

So $[Z_0] = \frac{V_{BR}}{I_{BR}}$

So, $\omega_{BR} = 3 I_{BR} R_0$

So $R_0 = \frac{\omega_{BR}}{3 I_{BR}}$

$\therefore [X_0] = \sqrt{[Z_0]^2 - [R_0]^2}$

$\Rightarrow \text{Efficiency } (\eta) =$

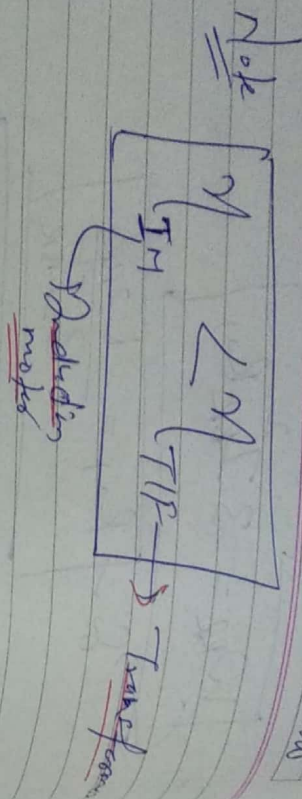
$\eta = \frac{\text{O/P Power}}{\text{I/P Power}}$

$\eta = \frac{P_{\text{out}} (\text{or}) P_{\text{rec}}}{P_{\text{in}} + \text{losses}}$

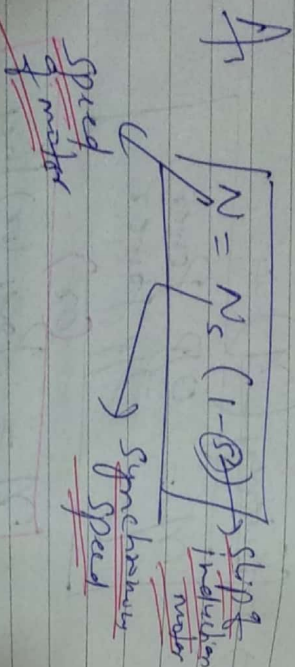
$\eta = \frac{P_{\text{out}}}{P_{\text{in}} + P_{\text{cu}} + P_{\text{fe}}}$

Calculated from blocked rotors

Inductively both mechanical loss



⇒ Speed control methods of induction motor:



⇒ Purpose of controlling speed of induction motor:

- To operate the loads smoothly
- To save the amount of power

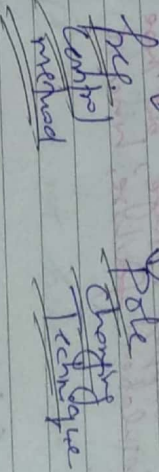
As $P \propto N$

$P = T \omega \Rightarrow T \propto \frac{2\pi N}{60}$
 $\Rightarrow P \propto N$

So, $P \propto N$

Speed control methods of induction motor:

- Slip control methods
- Synchronous speed control methods



$N_s = \frac{120f}{P}$
 Synchronous speed

① Slip control methods:

a) Voltage control method:

$T \propto \frac{s E_2^2}{R_2} \propto \frac{s V_1^2}{R_2}$
 Induction motor

$N = N_s (1 - s)$
 Synchronous speed

So, only $N < N_s$ is possible in this method

In any motor if voltage increases then it may cause insulation failure or damage. So in order to increase the voltage level in alternator or synchronous motor, level

2. Protection must be proper.

3. When the voltage reduces from the supply, slip of induction motor should increase in order to maintain the load torque constant. Now the speed of motor reduces.

Note for above rated speeds, it is required to increase the supply voltage, but this is a chance of insulation failure, hence it is not preferred.

⑥ Rotor resistance control method -

It is suitable for only slip ring induction motor. Since in squirrel cage induction motor it is not possible to vary rotor resistance.

$$T \propto \frac{1}{R_2} \quad \text{Control } [R_2 + R_e]$$

$$s \propto \frac{1}{N} = \frac{N_s(1-s)}{N}$$

So only $N < N_{rated}$ below rated speed is possible.

When on external resistance is connected in series with the rotor resistance, the rotor resultant impedance increases, but the voltage across the rotor drops. Hence, the gross torque of induction motor should increase. Now the speed of motor reduces.

Drawbacks of Rotor Resistance Control method

① It is not suitable for above rated speed.

② It is not suitable for starting.

③ Due to addition of resistance, the copper loss & rotor increases which reduces efficiency of motor.

④ Field injecting control -

$$T \propto \frac{s E_b^2}{R_2}$$

As the field into rotor can be injected in two ways

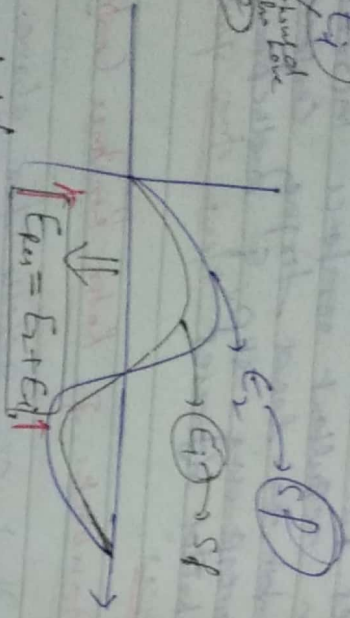
- 1) Three phase injecting method.
- 2) Out of phase injecting method.

mean (displacement)

1) Amplitude Injection method :-

Suppose this is added to rotor circuit

per phase
freq. ϕ



So, to add/inject an external emf E_2 in freq. should be same as that of slip frequency (ϕ) otherwise it is not possible to add two signals.

$$T \propto \frac{1}{s(E_2 + E_1)^2}$$

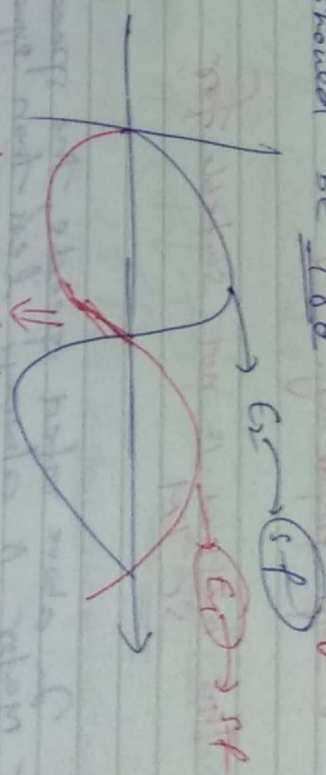
Hence

$$N = N_s (1-s)$$

$$s > N/N_{rated} \text{ above rated speed is possible}$$

2) Out of phase Injection control :-

In this method, the phase displacement b/w rotor emf E_2 & Injecting emf E_1 should be 180°



$$T \propto \frac{1}{s(E_2 - E_1)^2}$$

$$N = N_s (1-s)$$

$$N < N_{rated} \text{ So here speed possible is below rated.}$$

As Direct (First-Order method) are

1) It is difficult to produce a voltage slip frequency equal to ω .

2) This method is not suitable for SCIM.

Note In above rated speeds the speed of motor is always less than synchronous speed only.

→ Rated speed is the speed of motor at rated current.

② Synchronous Speed Control methods
a) Frequency Control methods

$$N_s = \frac{120f}{P}$$

$$N = N_s(1-s)$$

$$(N > N_{rated})$$

$$N_s = \frac{120f}{P}$$

$$N = N_s(1-s)$$

$$s, (N < N_{rated})$$

Suppose if $(f \downarrow)$ if freq. decreased so core get saturated.

$\rightarrow B_m \propto \frac{V}{f} \rightarrow$ so this ratio becomes constant

→ if set saturated

→ More rated speed

$$N_s = \frac{120f}{P}, N = N_s(1-s)$$

$$(N > N_{rated})$$

So, when frequency of supply increases, the N_s of the motor will also increase. The ratio to which the speed of motor also increases.

During this speed control, the B_m (reduces)

$$B_m \propto \frac{V}{f}$$

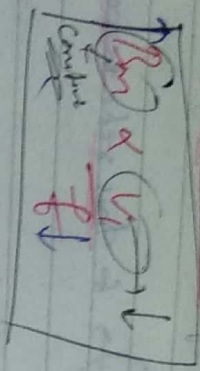
hence this method is called the "Pure frequency controlled method".

⇒ Below Rated Speed :-

$$I_m = \frac{120 \text{ fL}}{P}$$

$$I_m = I_{ms} (1-s)$$

$$N < N_{rated}$$



When the fep. of supply reduces the speed of motor will also reduce simultaneously but during this the flux density of core increases, hence avoid this saturation reduce the voltage also such that $\frac{V}{f} = \text{constant}$

This method is called Variable voltage & variable frequency control method

$$I \propto (N_s - N) \left(\frac{V}{f} \right)^2$$

ii) If $\frac{V}{f} = \text{constant}$ then $(N_s - N) \neq \text{constant}$

ii) If $\frac{V}{f} \neq \text{constant}$, so $(N_s - N) \left(\frac{V}{f} \right)^2 \rightarrow \text{constant}$

$$\frac{\text{Total speed}}{\text{constant}}$$

A 3- ϕ 440V, 50Hz SCIM driving a load at a speed of 960 rpm. If the voltage & frequency are reduced to 380V, 37.5Hz, then the speed of motor in order to maintain the constant load torque is _____?

Given 440V, 50Hz $\rightarrow N_1 = 960 \text{ rpm}$
 $\Rightarrow N_2 = 1000 \text{ rpm}$
Poles = 6

$$N_s = \frac{120 \times f}{P} = \frac{120 \times 50}{6} = 1000 \text{ rpm}$$

$$N_1, N_s, N_2 = N_s - N_2$$

$$\Rightarrow 1000 - 960 = \left[\frac{120 \times 37.5}{6} \right] - N_2$$

$$\Rightarrow N_2 = 950 \text{ rpm}$$

b) Pole changing method :-

as have better performance, design the stator poles.

As

$$N_s = 120f$$

P

$$N = N_s(1-s)$$

if $P=4$ Now if poles $\uparrow$ to 6

then $N_s \downarrow$ hence $N \downarrow$

So, when the poles of induction motor increases so speed $\downarrow$

When the poles of induction motor decreases so speed $\uparrow$

Note here, In this method, the speed obtain are the fixed speeds. So, it is not possible to get the smooth variations of speed.

This method suitable only to SCIM. Since the poles in SCIM are induced poles, but in case of SPIM the rotor poles are

designed for fixed ones. Hence, with poles changes in stator poles, the rotor poles will not change.

Note Mostly we prefer voltage control (or) frequency control methods.

A 3- ϕ , 6 pole 50Hz induction motor is operating with a rotor resistance of 0.2 Ω phase & running at 950 rpm. If it is required to operate at 800 rpm find the external resistance required to add in the rotor circuit is

Given No. of poles = 6
 $N_f = 950$ rpm
 $R_2 = 0.2 \Omega$
 $N_2 = 800$ rpm
 $R_{ext} = ?$

As $T \propto \frac{s}{R_2 + R_{ext}}$ in Rotor resistance method

$$N = N_s(1-s)$$

$$\frac{s}{R_2} = \text{constant} \quad \text{--- (1)}$$

$$N_s = 120 \frac{f}{P} = \frac{120 \times 50}{6} = 1000 \text{ rpm}$$

$$S_1 = \frac{N_s - N_1}{N_s}$$

$$S_1 = \frac{1000 - 950}{1000} = 0.05$$

$$\text{By } S_2 = \frac{N_s - N_2}{N_s} = \frac{1000 - 800}{1000} = 0.2$$

So from eq (1) we can write -

$$\sqrt{\frac{S_1}{R_1}} = \sqrt{\frac{S_2}{R_2 + R_{ex}}}$$

$$\Rightarrow \frac{0.05}{0.2} = \frac{0.2}{0.2 + R_{ex}}$$

$$\Rightarrow R_{ex} = 0.6 \Omega/\text{ph} \quad \underline{\text{Ans}}$$

2. The rotor emf of a 3- ϕ SLM is 200V running at 950 rpm. If an emf of 20V is injected in out of phase with the rotor emf then the speed of motor to maintain the constant load torque is

Soln Given

$$E_2 = 200V$$

$$N_1 = 950 \text{ rpm}$$

$$E_1 = 20V \rightarrow \text{out of phase}$$

then $N_2 = ?$

So, Direct field injecting method

$$I_r \propto S E_2$$

$$S_1 E_2^2 = \text{constant for motor}$$

to get

$$S_1 E_2^2 = S_2 (E_2 - E_1)^2$$

$$S_1 = \frac{1000 - 950}{1000} = 0.05$$

Assume $N_2 = 1000 \text{ rpm}$
 Let rotor emf to speed of motor

$$0.05 [200]^2 = S_2 [200 - 20]^2$$

$$S_2 = 0.06$$

$$N_2 = N_s (1 - S)$$

$$N_2 = 1000 (1 - 0.06)$$

$$N_2 = 940 \text{ rpm} \quad \underline{\text{Ans}}$$

Q. In a 3φ induction motor, if the supply voltage is reduced by 20% then the % torque is decreased by

a) 20% b) 40% c) 36% d) 10%

Sol:

$$T \propto \frac{180}{\pi N_s} \left\{ \frac{s E_1^2 R_2}{R_2^2 + (s X_2)^2} \right\}$$

$$R_2, T \propto E_1^2 \propto V_1^2$$

$$8 \text{ } \downarrow \text{ } V \downarrow \text{ by } 20\% \Rightarrow T \downarrow$$

$$\text{Support } T \propto 1 \cdot V^2$$

ie $T \propto (0.8)^2$

$$T \propto 0.64$$

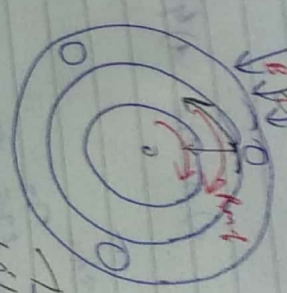
$$0.36 \downarrow$$

ie T is reduced by 36%

Q. In a 3φ IM is operating at slip s. If its slip but that instant is

a) s b) 1-s c) 2-s d) 1+s

Sol:



So, when two leads are interchanged then the direction of rotation will reverse instantly, but not the direction of rotor

$$\text{Relative speed} = N_s + N$$

$$\text{in backward slip} = (s_b) = (1-s)$$

$$s' = \frac{N_s - N}{N_s} = \frac{1-s}{1}$$

ie $s' = 1-s$

Q 4-4 pole induction motor supplied by a slightly unbalanced 3-ph 50Hz supply is added at 1440 rpm. The frequency is _____ sequence current in the motor is _____

- a) 165 Hz b) 98 Hz c) 52 Hz d) 98 Hz
 e) 52 Hz f) 98 Hz

Given pole = 4, $f_s = 50$ Hz
 $N_s = \frac{120f}{p} = \frac{120 \times 50}{4} = 1500$ rpm

Not advantages from is an unbalanced like 3ph supply then the backward (or) -ve sequence current are possible
 $s = \frac{N_s - N}{N_s} = \frac{1500 - 1440}{1500} = \frac{60}{1500} = 0.04$

Advantages from is an unbalanced like 3ph supply then the backward (or) -ve sequence current are possible

At 1500 rpm Forward slip
 $s_p = 0 \Rightarrow f_{2r} = 5$

Backward slip
 $s_b = 2 - 5 \Rightarrow f_{2r} = (2.5)f$

$f_{2r} = (2 - 5) \cdot f$
 $= (2 - 0.04) \times 50 = 98 \text{ Hz}$

Q The locked rotor current in 3-ph-4 connected, 15kW, 4-pole, 230V, 50Hz IM. at rated condition is 50A. Neglecting the core magnetizing current, the approximate locked rotor line current drawn when the motor is connected to 236V, 57Hz supply is _____

- a) 58.5 A b) 45 A c) 45.7 A d) 55.5 A

Given $I_1 = 50$ A
 230 V, 50 Hz, $p = 4$

Power rating $P = 15 \text{ kW}$
 $I_2 = ?$
 236 V, 57 Hz

8, 11 WKT $I_2 = \frac{P_2}{\sqrt{I_1^2 + (X_1)^2}}$

Since blocked rotor $s = 1$, so

$I_2 = \frac{P_2}{\sqrt{R_2^2 + (X_2)^2}}$

Long neglected so $k_2 = 0$

$I_2 = \frac{P_2}{\sqrt{R_2^2 + (X_2)^2}}$

$I_2 \propto \frac{V}{\sqrt{R_2^2 + (X_2)^2}}$

So, $I \propto \frac{V}{f}$

here $\frac{T_2}{T_1} = \frac{V_2}{V_1} \times \frac{f_1}{f_2} \Rightarrow T_2 = \frac{236}{236} \times \frac{50}{54}$

$\Rightarrow T_2 = 4.54 \text{ Nm}$

A 3- ϕ , 440V, 1042 IM is running at rated speed with a fixed load. If the supply voltage is reduced by half (1/2) frequency but developed in motor is _____

Ans a) reduced by 1/2

b) remain same

c) increased by 1/2

d) doubles

Given $T_{max} = \frac{180}{2\pi N_s} \left\{ \frac{E_2^2}{2X_2} \right\}$

$\Rightarrow T_{max} \propto \frac{V^2}{f^2} \propto \left(\frac{V_1}{f_1} \right)^2 = \left(\frac{V_2}{f_2} \right)^2$

Since $T_{max} = \text{constant}$ (or) remain same

Support By keeping $V = \text{constant}$, if supply freq. is double then T_{max} is _____

So, $T_{max} \propto \left(\frac{V_1}{f} \right)^2$

So if freq. doubles

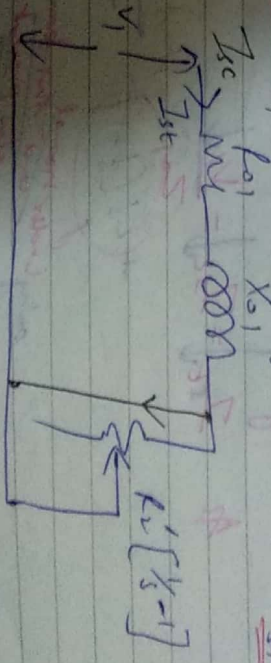
So $T_{max} \propto \left(\frac{1}{2} \right)^2$

So T_{max} reduced by 1/4

$\Rightarrow$ Starting methods of 3- ϕ Induction Motor

Starting methods are required to limit the high starting current so that windings of motor may not get damaged.

Consider equivalent circuit of IM referred to stator.



at the time of starting means $(N-1) \times I_{sc}$

$$S = \frac{N-1}{N} \Rightarrow (S=1)$$

So, Replace $S=1$ in $R_2' \left[\frac{1}{S}-1 \right]$

By applying KVL $\Rightarrow V_1 = I_{sc} [R_{01} + jX_{01}]$

$$\Rightarrow V_1 = I_{sc} Z_{01}$$

$I_{sc} \Rightarrow$ starting current $\rightarrow 5\% \text{ to } 8\% \text{ of } I_{fe}$

$$\left[\frac{I_{sc}}{Z_{01}} = \frac{V_1}{Z_{01}} \right]$$

So, because of this high current the winding will damage.

As $I_{sc} = I_{sc} = \frac{V_1}{Z_{01}}$

As Z_{01} is fixed $\Rightarrow \frac{V_1}{Z_{01}}$

only under and also protected

only possibility to reduce the supply

voltage to limit the starting current.
So, to order to protect the winding of IM from the high starting currents, it is preferred to reduce the supply voltage at the time of starting.

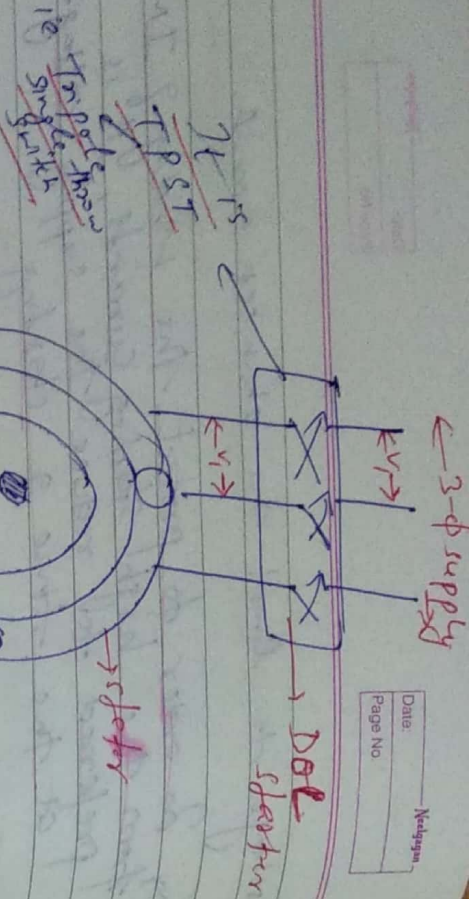
Starting Methods of IM

- 1) Direct starting (Direct on line starting)
- 2) Series reactor starting
- 3) Series resistor starting
- 4) Auto-T/F starting $\rightarrow$ preferred in large as a variable
- 5) Δ starting method $\rightarrow$ preferred for domestic purpose only this method is in general starting method.

1) DOL starting method:

$$\left[\frac{I_{sc}}{I_{pr}} \right] \propto \left[\frac{I_{sc}}{I_{pr}} \right]^2$$

As I_{sc} is fixed $\Rightarrow \left[\frac{I_{sc}}{I_{pr}} \right]^2$



Starting current

$$I_{st} = \frac{V}{Z_{01}} \quad \gamma = 7$$

eg, $I_{st} = 7 I_{sc}$

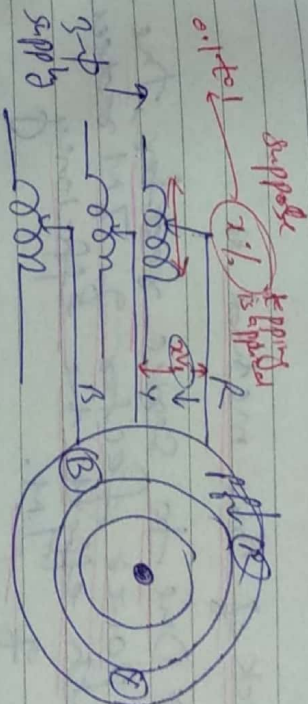
Hence, In DOL starting method, once the starter is operated then the total supply voltage will appear across the starting current drawn in this method is same as that of Short circuit current i.e. (I_{sc})

$$\frac{I_{st}}{I_{fl}} = \left(\frac{I_{sc}}{I_{fl}} \right)^2 \cdot 5fl$$

eg, In this method there is no limit on short circuit current but it is considered as reference

Note: eg, This method is taken as reference to compare with other methods

2. Series Reactor starting method :-
Here we place reactor in series with motor.



If $x\%$ tapping is applied on series reactor. Then, the voltage across the reactor appeared is also $x\%$ which is reduced by fraction x . Hence the $I_{st} = \frac{V}{Z_{01}} \cdot \frac{1}{x}$

$$I_{st} = x [I_{sc}]$$

Compared to DOL method the I_{st} drawn in this method is reduced by a fraction x & I_{sc}

$$\frac{I_{st}}{I_{fl}} = \left(\frac{x I_{sc}}{I_{fl}} \right)^2 \cdot 5fl$$

$$\frac{I_{sc}}{I_{pr}} = x^2 \left[\frac{I_{sc}}{I_{pr}} \right]^2 \text{ s.p.e.}$$

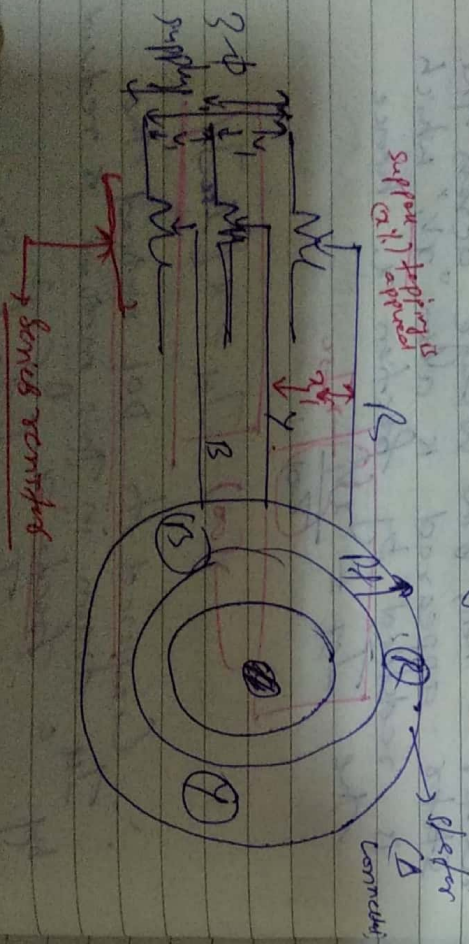
So, starting voltage also reduced by fraction of x^2 .

So, compared to DOL method, the starting torque will also reduce by $\frac{1}{x^2}$.

Drawback of this method is

→ Due to series reactor, the power factor of IM reduces which effect the efficiency of IM.

③ Series resistor starting method :-



$$I_{sc} = \frac{V}{Z_{01}} + I_{sc}$$

$$I_{sc} = \frac{V}{Z_{01} + x Z_{sc}}$$

$$\frac{I_{sc}}{I_{pr}} = \left(\frac{I_{sc}}{I_{pr}} \right)^2 \text{ s.p.e.}$$

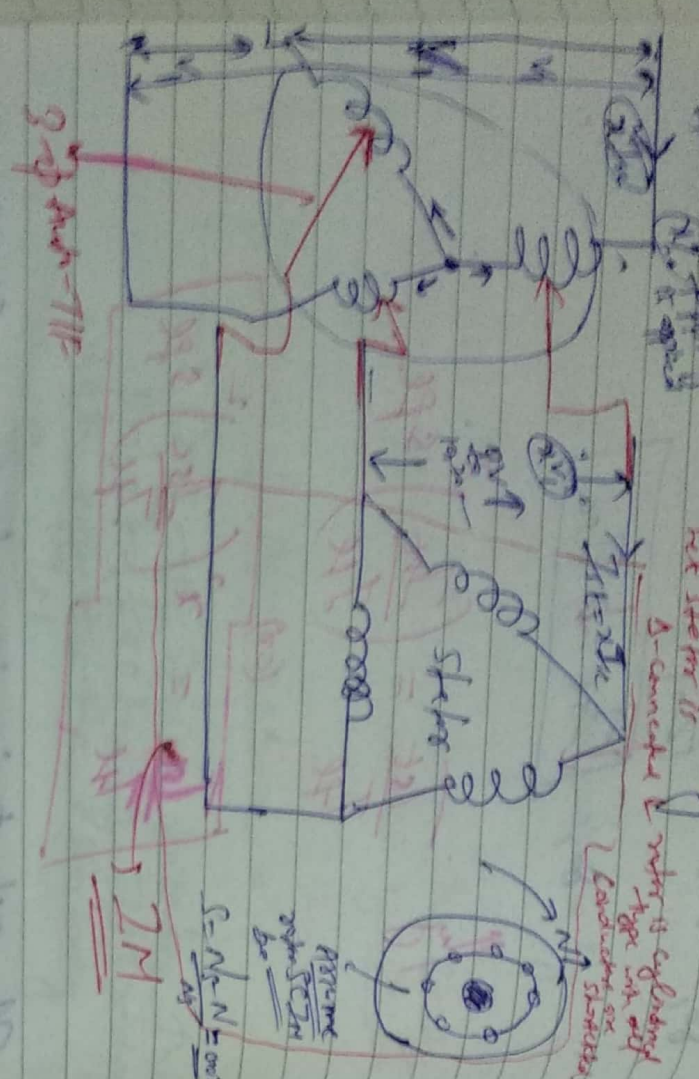
$$\frac{I_{sc}}{I_{pr}} = x^2 \left(\frac{I_{sc}}{I_{pr}} \right)^2 \text{ s.p.e.}$$

Q. x% tapping is applied on series resistors then the starting current reduces by a fraction x , the starting torque reduces by a fraction x^2 compared to DOL method.

→ The power factor of IM improves by replacing the reactor with resistor.

Drawback :-
→ Due to presence of series resistor the copper loss increases which reduces the efficiency of motor.

4) Auto-Transformer starting :-



Let stator is Δ-connected & motor is star connected. Hence, the supply is reduced by a factor α .

If 24 tapping is applied on an Auto-TTF, then the current drawn by TM reduces by a factor α compared to the full method.

At the same time the current drawn by auto primary & auto-TTF from supply is reduced by a factor α .

$$I_{st} = \left[\frac{I_t}{\alpha} \right] \times 5H$$

$$= \left[\frac{1 \times I_{sc}}{1} \right] \times 5H$$

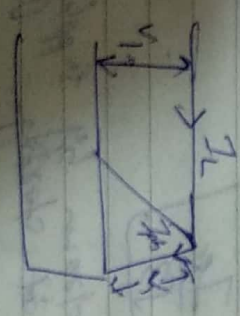
$$\left[\frac{I_t}{\alpha} \right] = \alpha^2 \left(\frac{I_{sc}}{1} \right) \times 5H$$

Compared to other methods for the same reduction in torque, the current drawn from supply is reduced by factor α .

Disadvantage is $\rightarrow$ The size cost & complexity is more in circuit designing.

5) Y/Δ starting :-

Assume $\Delta \rightarrow$ Connected stator



Let 240V be phase voltage

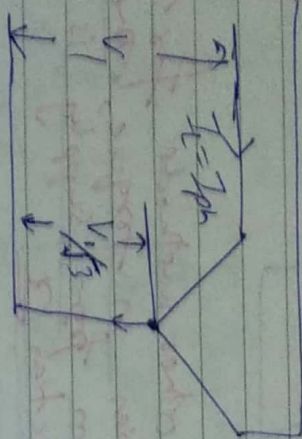
$$V_L = \frac{V}{\sqrt{3}}$$

$$I_L = \sqrt{3} I_{ph}$$

$$I_L = \sqrt{3} I_{sc}$$

Q3 how current is less by 1/3 at starting of motor when it is connected in star supply is 240V.

Assume Y-connected stator



$$I_{ph} = \frac{V_L / \sqrt{3}}{Z_{01}}$$

$$I_{ph} = \frac{1}{\sqrt{3}} \left(\frac{V_L}{Z_{01}} \right) \rightarrow I_{sc}$$

$$I_{ph} = \frac{I_{sc}}{\sqrt{3}}$$

$$I_L = I_{ph} = \frac{I_{sc}}{\sqrt{3}}$$

When current drawn by supply is reduced at starting.

At starting Y connected is preferred but after starting i.e. in normal running condition Δ connection is preferred.

This starting method is possible only when there is a provision of both star (Y) & delta (Δ) connected winding is present in the IM.
Since in case of Y connected stator, the starting current drawn is $\frac{1}{\sqrt{3}}$ times of I_{sc} which is limited without any external device.

Hence Y-connected stator is preferred at starting & Δ -connected stator is preferred under running condition.

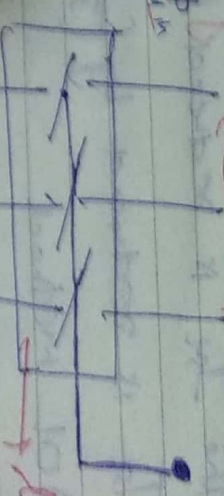
The change of connection from (Y to Δ) can be made at starting by using (Y/ Δ -Switch) (Stator-Y)

Let above motor connected in

at starting

at then 3- ϕ connected

stator- Δ



like two way switch

$$\underline{Y} = \underline{I}_{st} = \frac{I_{sc}}{3}$$

$$\underline{I}_{sc} = \left[\frac{I_{sc}}{I_{sc}} \right]^2 S_{sc}$$

$$= \left[\frac{I_{sc}/13}{I_{sc}} \right]^2 S_{sc}$$

$$\underline{I}_{sc} = \frac{1}{3} \left[\frac{I_{sc}}{I_{sc}} \right]^2 S_{sc}$$

Compare by DOL starting method
the I_{st} is reduced by $\left(\frac{1}{3}\right)$ times

I_{st} is reduced by $\left(\frac{1}{3}\right)$ times

DOL method

At Auto-7/E starting

$$\underline{I}_{st} = x \cdot I_{sc} \quad \text{--- (1)}$$

But for Y/A starting

$$\underline{I}_{st} = \frac{I_{sc}}{3} \quad \text{--- (2)}$$

So, comparing to (1) & (2) we have

$$\frac{x \cdot I_{sc}}{3} = 0.5 I_{sc}$$

At 57.7% of tapping current
Auto-7/E, the starting current

drawn by Y/A starting & Auto-7/E
starting are same

The starting current of a 3- ϕ TM is
5 times the rated current, while the
rated slip is $\frac{1}{4}$, the ratio of

$$\underline{I}_{sc} = 2$$

- a) 0.8 b) 0.6 c) 1.2 d) 1

So, A correct

$$\underline{I}_{st} = \left(\frac{I_{sc}}{I_{sc}} \right)^2 S_{sc}$$

Given, $\underline{I}_{st} = 5 I_{sc}$ & $S_{sc} = \frac{1}{4} = 0.25$

$$\underline{I}_{sc} = \left(\frac{5 I_{sc}}{I_{sc}} \right)^2 \times 0.25 = 1 \underline{I_{sc}}$$

Q. The suitable tapping on an auto-tap when the T.M. required to start with 40% of full load torque. The short ckt current drawn is 5 times the full load current & full load slip is 0.035 is _____?

- a) 57.7%
 b) 67.6%
 c) 66.7%
 d) 60%

Given At WKT $\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st}}{I_{fl}} \right)^2 \times s_{fl}$

But in case of Auto-TIF

$\frac{T_{st}}{T_{fl}} = x^2 \left[\frac{I_{st}}{I_{fl}} \right]^2 s_{fl}$

2. Given $T_{st} = 40\% \text{ of } T_{fl}$

$I_{st} = 5 I_{fl}$

$s_{fl} = 0.035 = 3.5\%$

So eq ① becomes

$0.4 \frac{T_{fl}}{T_{fl}} = x^2 \left(\frac{5 I_{fl}}{I_{fl}} \right)^2 \times 0.035$

$x = 0.676 = 67.6\%$